Structures and Materials

Dr.-Ing. Andreas Hermanutz

Can equilibrium be maintained?

Topics:

How do we simplify it?

Introduction/fundamental tools → **Elasticity** → **Structural Idealisations** → **Material Behaviour** → **Stability** → **Numerical Structural Analysis.**

How does a deformable body behave?

How do we solve complex structures?

What behaviour and failure do we assign?

Topic 1: Engineering tools and concepts

Topic 2: Loads, stress and deformation

How do we describe the behaviour of a deformable body?

- equilibrium → stress
- kinematics → strain/deformation
- constitutive law → connection between stress and strain.

- **Stress is a tensor:** the stress state contains normal and shear stresses and depends on the orientation of the considered plane.
- **Strain measures deformation**, while displacement describes the movement of material points; rigid-body motion does not create strain.

A deformable structure is described by equilibrium, kinematics and a material law. Together they connect loads, stresses and deformation.

This continuum description is general, but solving the complete 3-D elasticity problem for every structure would be impractical. We therefore introduce structural idealisations.

Topic 3: Structural elements

How do we simplify the continuum into useful engineering models?

In Topic 2 we described a general elastic continuum. We now reduce this theory to 1-D and 2-D structural elements by introducing assumptions about geometry and deformation.

- Trusses/bars: mainly axial loads.
- Beams: bending, shear and torsion.
- Membranes: mainly in-plane loads.
- Plates: mainly out-of-plane loads.
- Shells: combine membrane and plate behaviour.

Structural stiffness combines material and geometry:
material property × geometrical property (EA, EI, GIt, GA)

Structural elements make the problem solvable by replacing the general continuum with simplified models — but every model is only valid within its assumptions.

Topic 4: Material selection

What material behaviour do we put into our structural model?

Material properties are part of the structural model: choose the material according to the required behaviour, and use the constitutive and failure model appropriate to that material.

- Material selection starts from structural and operational requirements, not from a single material property.
- Relevant properties include density, stiffness, strength, ductility/toughness, fatigue, corrosion and thermal behaviour, as well as manufacturing and cost.
- Metals: largely isotropic structural behaviour; strength must be assessed with an appropriate failure/yield criterion.
- Composites: directional/orthotropic behaviour; stiffness and strength can be tailored through fibre orientation and laminate lay-up.
- Material selection therefore affects both the constitutive model and the failure model used in structural analysis.

Topic 5: Structural stability and buckling

When is strength no longer the governing design criterion?

A structure can lose stability before it reaches its material strength; therefore lightweight structures must be checked for critical buckling and post-buckling behaviour.

- Buckling occurs particularly in slender and thin-walled structures under compression or shear.
- At the critical load, the original equilibrium state loses stability and a new deformed equilibrium configuration becomes possible.
- Geometry, stiffness, boundary conditions and imperfections strongly influence the critical buckling load.
- Distinguish local and global buckling.
- Buckling does not necessarily mean immediate collapse: post-buckling behaviour can provide additional load-carrying capability.

equilibrium stability → elastic column → imperfections → plates → practical buckling assessment and post-buckling.

For lightweight structures, design is not only a question of stress and strength; structural stability can become the governing limitation.

Topic 6: Computational structural analysis (FEM)

How do we apply all previous concepts to complex structures?

- **Discretisation**: divide the continuous structure into finite elements connected by nodes and degrees of freedom.
- Each element contributes its **stiffness**, based on its geometry, element formulation and material law.
- Transform and **assemble the element equations** into the global structural system.
- Apply **loads and boundary conditions** and solve the system.
- Recover displacements → strains → stresses and assess the structural result.
- FEM does not replace the mechanics from Topics 1–5; it is the numerical implementation of those mechanics.

FEM allows us to solve complex structural systems numerically, but the quality of the answer still depends on the structural model, material assumptions, element choice and boundary conditions.

1 Materials	(/ 4.0)

a) Which statement about *rigid body motions* is correct?

- [ ] They produce stress but no strain
- [x] They produce neither stress nor strain
- [ ] They always produce strain
- [ ] They produce no stress, but produce strain
- [ ] None is correct

b) For an isotropic linear elastic material, how many independent elastic constants exist?

- [ ] 21
- [ ] 81
- [ ] 3
- [ ] 6
- [x] None is correct

c) In an orthotropic material, the anisotropic constitutive matrix becomes simpler because:

- [ ] Shear stresses are always zero.
- [ ] Poisson's ratio is always zero.
- [x] Many coupling terms vanish due to symmetry planes.
- [ ] The material behaves identically in all directions
- [ ] None is correct

d) The Green strain tensor is used instead of the small strain tensor when:

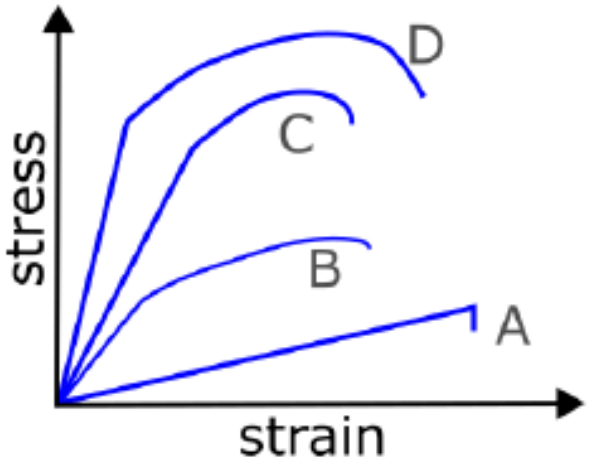
- [ ] Stresses are small.
- [ ] Material is brittle.
- [x] Deformations are large or include large rotations.
- [ ] Poisson's ratio is unknown.
- [ ] None is correct

2 Materials	(/ 4.0)

a) State in the table below typical values for the Young's modulus of the listed materials in MPa.

b) Identify the corresponding qualitative tensile test curves **A-D** for the materials listed in part a).

Material	Young's Modul [MPa]	Tensile-test Curve
Aluminium	70000	B
Steel	210000	D
Titanium	110000	C
Quasi-Isotropic GFRP	24000	A



3 Composites	(/ 1.0)

a) For a unidirectional (UD) composite ply loaded in the fibre direction, which statement is most accurate?

- [x] Increasing fibre volume fraction generally increases the modulus in fibre direction.
- [ ] The stiffness in fibre direction is dominated by the matrix modulus.
- [ ] The stiffness in fibre direction is independent of fibre properties.
- [ ] The modulus in fibre direction is always equal to the matrix modulus.
- [ ] None is correct

4 Buckling	(/ 4.0)

a) Explain the difference between global and local buckling in lightweight structures.

Global buckling: the whole structural member bends or deflects as one unit. The global load-carrying capacity may be exceeded, which can lead to collapse.

Local buckling: only a smaller part of the structure, such as a plate, flange, or web, buckles while the overall member remains mostly straight. The load ca/might redistribute, so the structure may remain stable and the deformation can be reversible if the yield strength is not exceeded.

4 Buckling	(/ 4.0)

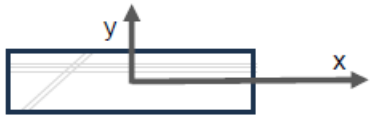
b) The critical buckling load of the panel is $P_{crit} = 10.0\ kN$. Due to a new manufacturing method, the material strength σ is reduced by 10 %. What is the new critical load $P_{crit,new}$?

No change. Elastic buckling is stiffness-dependent, not strength-dependent; therefore Pcrit,new=10.0kN.

5 Classical Laminate Theory	(/ 10.0)

Given a $[0/45/0]_S$ CFRP laminate with each layer having a thickness of 0.25mm. Determine the **A11, A12** and **B11** coefficients of the laminate stiffness matrix.

Material properties: $E_1 = 138\ [GPa]$, $E_2 = 8960\ [MPa]$, $G_{12} = 7.1\ [GPa]$, $\nu_{12} = 0.3$



a) Calculate the plane stiffness matrix for the 0° and 45° ply.

$$\nu_{21} = \nu_{12}\frac{E_2}{E_1} = 0.3\frac{8.96}{138} = 0.01948$$

$$Q_{11} = \frac{E_1}{1-\nu_{12}\nu_{21}} = 138.811\ \text{GPa}$$

$$Q_{22} = \frac{E_2}{1-\nu_{12}\nu_{21}} = 9.013\ \text{GPa}$$

$$Q_{12} = \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} = 2.704\ \text{GPa}$$

$$Q_{66} = G_{12} = 7.1\ \text{GPa}$$

$$\bar{Q}^{0^\circ} = \begin{bmatrix} 138.811 & 2.704 & 0 \\ 2.704 & 9.013 & 0 \\ 0 & 0 & 7.100 \end{bmatrix}\ \text{GPa}$$

$$m = \cos 45^\circ = \frac{1}{\sqrt{2}}, \qquad n = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$T = \begin{bmatrix} m^2 & n^2 & 2mn \\ n^2 & m^2 & -2mn \\ -mn & mn & m^2-n^2 \end{bmatrix} \qquad T_{45} = \begin{bmatrix} 0.5 & 0.5 & 1 \\ 0.5 & 0.5 & -1 \\ -0.5 & 0.5 & 0 \end{bmatrix}$$

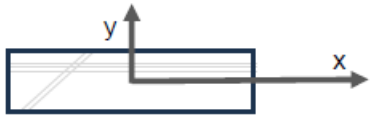
$$\bar{Q} = T^{-1}QT$$

$$\bar{\mathbf{Q}}_{45^\circ} = \begin{bmatrix} 45.408 & 31.208 & 32.450 \\ 31.208 & 45.408 & 32.450 \\ 32.450 & 32.450 & 35.604 \end{bmatrix}\ \text{GPa}$$

5 Classical Laminate Theory	(/ 10.0)

Given a $[0/45/0]_S$ CFRP laminate with each layer having a thickness of 0.25mm. Determine the **A11, A12** and **B11** coefficients of the laminate stiffness matrix.

Material properties: $E_1 = 138\ [GPa]$, $E_2 = 8960\ [MPa]$, $G_{12} = 7.1\ [GPa]$, $\nu_{12} = 0.3$



b) Determine the A11, A12 and B11 coefficients of the laminate stiffness matrix.

$$A_{ij} = \sum_k \bar{Q}_{ij}^{(k)} t_k$$

$$A_{11} = 0.25\,[4(138811.14) + 2(45407.85)]$$
$$A_{11} = 161515\ \mathrm{N/mm}$$

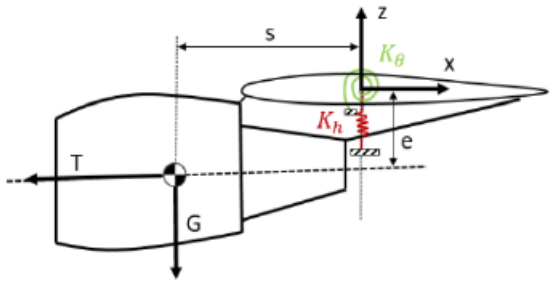
$$A_{12} = 0.25\,[4(2703.80) + 2(31207.85)]$$
$$A_{12} = 18308\ \mathrm{N/mm}$$

symmetric laminate $\Rightarrow [B] = 0$

6 Structural Dynamics	(/ 12.0)

Given is an engine, mounted on a wing, as shown in the figure below. The equations for a 2-DoF system shall be formulated to evaluate the structural dynamics. Two forces, thrust T and weight G, shall be considered. Aerodynamic forces can be neglected. The engine shall be assumed to be concentrated point mass. The bending stiffness is modelled with a spring stiffness K_h and the torsional stiffness is modelled with the rotational spring stiffness K_Θ.





$K_h = 550\frac{\text{kN}}{\text{m}}$, $K_\Theta = 4200\frac{\text{kNm}}{\text{rad}}$, $m_{engine} = 6160\ \text{kg}$, $e = 1.5\ \text{m}$, $s = 3.2\ \text{m}$

a) Formulate the linearized equations of motion for the two coordinates h (translation in z direction) and Θ (rotation around y) in the shown coordinate center. Potential energy from the location of the center of gravity outside the coordinate center shall be neglected.

$$\dot{z}_G = \dot{h} + s\dot{\theta}, \qquad \dot{x}_G = -e\dot{\theta}$$

$$T_{\text{kin}} = \frac{1}{2}m\left[(\dot{h} + s\dot{\theta})^2 + (e\dot{\theta})^2\right]$$

$$T_{\text{kin}} = \frac{1}{2}m\dot{h}^2 + ms\dot{h}\dot{\theta} + \frac{1}{2}m(s^2 + e^2)\dot{\theta}^2$$

$$\mathbf{M} = \begin{bmatrix} m & ms \\ ms & m(s^2+e^2) \end{bmatrix}$$

$$\mathbf{M} = \begin{bmatrix} 6160 & 19712 \\ 19712 & 76938.4 \end{bmatrix}$$

$$V = \frac{1}{2}K_h h^2 + \frac{1}{2}K_\theta \theta^2$$

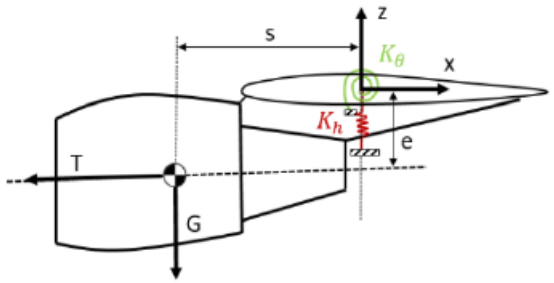
$$\mathbf{K} = \begin{bmatrix} K_h & 0 \\ 0 & K_\theta \end{bmatrix}$$

$$\begin{bmatrix} 6160 & 19712 \\ 19712 & 76938.4 \end{bmatrix}\begin{bmatrix} \ddot{h} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} 550000 & 0 \\ 0 & 4.2\times 10^6 \end{bmatrix}\begin{bmatrix} h \\ \theta \end{bmatrix} = \begin{bmatrix} -G \\ eT - sG \end{bmatrix}$$

6 Structural Dynamics	(/ 12.0)

Given is an engine, mounted on a wing, as shown in the figure below. The equations for a 2-DoF system shall be formulated to evaluate the structural dynamics. Two forces, thrust T and weight G, shall be considered. Aerodynamic forces can be neglected. The engine shall be assumed to be concentrated point mass. The bending stiffness is modelled with a spring stiffness K_h and the torsional stiffness is modelled with the rotational spring stiffness K_Θ.





$K_h = 550\frac{kN}{m}$, $K_\Theta = 4200\frac{kNm}{rad}$, $m_{engine} = 6160\ kg$, $e = 1.5\ m$, $s = 3.2\ m$

b) The system has now been transformed into modal coordinates and has been mass normalized. Calculate all the eigenfrequencies for the mounted engine system:
$[M_{G,n}]\{\ddot{q}\} + [K_{G,n}]\{q\} = \{Q_{G,n}\}$; $K_{G,n,11} = 35.45$; $K_{G,n,22} = 763.22$

$\omega_1 = \sqrt{35.45}$

$\omega_2 = \sqrt{763.22}$

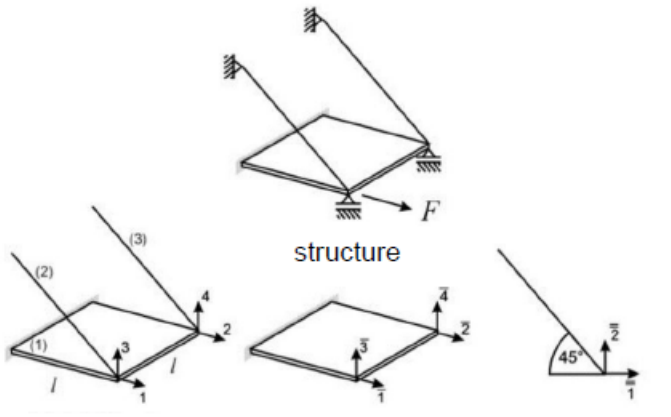
$f = \frac{\omega}{2\pi}$

$f_1 \approx 0.95\ Hz$

$f_2 \approx 4.40\ Hz$

7 FEM	(/ 10.0)

The following structure shall be examined by means of FEM. All stiffness matrices are already given in the rotated degrees of freedom as sketched below.



structure

global DOF and element numbers

plate stiffness

strut stiffness

Plate: $[K_P] = \begin{bmatrix} 15 & -10 & 0 & 0 \\ -10 & 15 & 0 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix}$ Strut: $[K_S] = \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix}$

a) Assemble the global stiffness matrix for all those DOFs where U is not zero.

Plate: $[K_P] = \begin{matrix} \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{bmatrix} 15 & -10 & 0 & 0 \\ -10 & 15 & 0 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix} \end{matrix}$

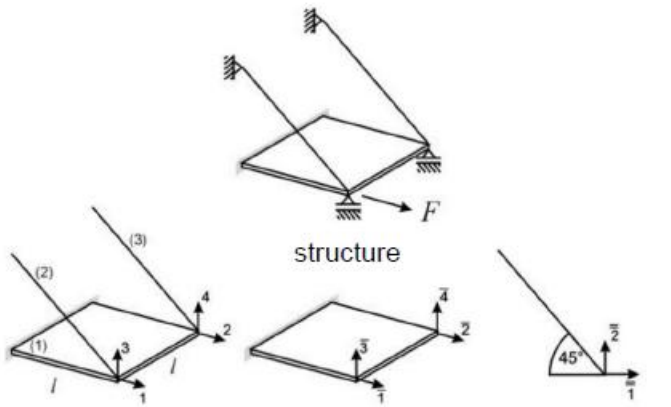
Strut: $[K_S] = \begin{matrix} \\ 1 \\ 3 \end{matrix} \begin{matrix} \begin{matrix} 1 & 3 \end{matrix} \\ \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix} \end{matrix}$

$[K_S] = \begin{matrix} \\ 2 \\ 4 \end{matrix} \begin{matrix} \begin{matrix} 2 & 4 \end{matrix} \\ \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix} \end{matrix}$

$$\mathbf{K}_G = \begin{matrix} \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{matrix} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{bmatrix} 23 & -10 & -4 & 0 \\ -10 & 23 & 0 & -4 \\ -4 & 0 & 13 & 2 \\ 0 & -4 & 2 & 13 \end{bmatrix} \end{matrix}$$

7 FEM	(/ 10.0)

The following structure shall be examined by means of FEM. All stiffness matrices are already given in the rotated degrees of freedom as sketched below.



global DOF and element numbers | plate stiffness | strut stiffness

Plate: $[K_P] = \begin{bmatrix} 15 & -10 & 0 & 0 \\ -10 & 15 & 0 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix}$ Strut: $[K_S] = \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix}$

b) Calculate the displacement U_1 and U_2 in relation to the force F. If necessary, you can use a backup result for a): $[K] = \begin{bmatrix} 20 & 4 \\ 4 & 20 \end{bmatrix}$

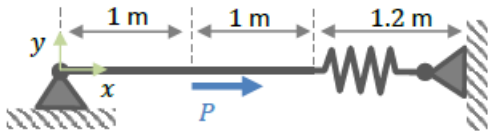
$$\begin{bmatrix} 20 & 4 \\ 4 & 20 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

$$U_1 = \frac{5}{96}F$$

$$U_2 = -\frac{1}{96}F$$

8 FEM	(/ 8.0)

A structure consists of a continuous truss connected to a discrete spring. At the position x=1m, a load of 3.5 kN is applied. The truss is made of aluminium, with a modulus of elasticity of 68 GPa and a cross-sectional area of 350 mm². The spring has a stiffness of 1200 kN/m.



a) Calculate the displacement at $x = 1$ m.

$$k_b = \frac{EA}{L} = \frac{68 \times 10^9 (350 \times 10^{-6})}{1} = 23.8 \times 10^6 \text{ N/m}$$

$$\begin{bmatrix} 2k_b & -k_b \\ -k_b & k_b + k_s \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} P \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 47600 & -23800 \\ -23800 & 25000 \end{bmatrix} \begin{bmatrix} u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} 3.5 \\ 0 \end{bmatrix}$$

$$u(x = 1\,\text{m}) \approx 0.140 \text{ mm}$$